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MultiDTI: drug–target interaction prediction based on multi-modal representation learning to bridge the gap between new chemical entities and known heterogeneous network

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Abstract

Motivation: Predicting new drug–target interactions is an important step in new drug development, understanding of its side effects and drug repositioning. Heterogeneous data sources can provide comprehensive information and different perspectives for drug–target interaction prediction. Thus, there have been many calculation methods relying on heterogeneous networks. Most of them use graph-related algorithms to characterize nodes in heterogeneous networks for predicting new drug–target interactions (DTI). However, these methods can only make predictions in known heterogeneous network datasets, and cannot support the prediction of new chemical entities outside the heterogeneous network, which hinder further drug discovery and development.

Results: To solve this problem, we proposed a multi-modal DTI prediction model named ‘MultiDTI’ which uses our proposed joint learning framework based on heterogeneous networks. It combines the interaction or association information of the heterogeneous network and the drug/target sequence information, and maps the drugs, targets, side effects and disease nodes in the heterogeneous network into a common space. In this way, ‘MultiDTI’ can map the new chemical entity to this learned common space based on the chemical structure of the new entity. That is, bridging the gap between new chemical entities and known heterogeneous network. Our model has strong predictive performance, and the area under the receiver operating characteristic curve of the model is 0.961 and the area under the precision recall curve is 0.947 with 10-fold cross validation. In addition, some predicted new DTIs have been confirmed by ChEMBL database. Our results indicate that ‘MultiDTI’ is a powerful and practical tool for predicting new DTI, which can promote the development of drug discovery or drug repositioning.

Availability and implementation: Python codes and dataset are available at <https://github.com/Deshan-Zhou/MultiDTI/>.

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Supplementary information: [Supplementary data](#) are available at *Bioinformatics* online.

1 Introduction

The identification of drug–target interactions (DTI) has attracted great attention because it is an important step in the development of new drugs, understanding of their side effects and drug repositioning (Ding *et al.*, 2014). Most DTI are formed by drugs competing with endogenous small molecules for functionally needed binding sites on

proteins (Hopkins and Groom, 2002), which in turn affects disease conditions. There are currently thousands of FDA-approved drugs on the market, as well as drugs in the late stage of clinical trials (Xia *et al.*, 2010). These drugs may have potential unobserved targets and unknown indications. However, it is time-consuming, laborious and expensive to determine DTIs through biological experiments (Zheng *et al.*, 2020). Therefore, there is a strong motivation to

establish a computational model that can estimate the interaction of new drug–target pairs and provide biologists with DTI candidates to reduce the workload of wet laboratory experiments.

The traditional computer-based methods for predicting DTI mainly include docking-based methods (Garrett, 2009) and ligand-based methods (Michael, 2007). Docking-based methods require the three-dimensional structure of the target protein. These methods have limitations because not all target structural information is known. The ligand-based methods use known rules of interacting ligands to make predictions. If the number of known ligands is insufficient, these methods will not work well.

In recent years, the methods of predicting DTI through calculation mainly include similarity-based methods and network-based methods. The similarity-based methods rely on the assumption that compounds with similar structures may have similar properties. An early method used a drug–drug and gene–gene similarity measures, combined with a logistic regression component to integrate the scores of multiple metrics to produce the final association score (Perlman et al., 2011). Early similarity-based methods mostly used hand-made features. In recent years, studies have shown that integrating feature learning capabilities into machine learning models can improve prediction performance (Abbasi et al., 2020). Recently, deep learning methods have received more and more attention because they seem to overcome certain limitations and reduce the loss of feature information in the prediction process (Maryam et al., 2020). ‘DeepDTIs’ (Wen et al., 2017) first automatically extracts the characteristics of drugs and targets from simple chemical substructure and sequence information, and then uses an effective deep learning method-Deep Belief Network (DBN) to construct a classification model. ‘DeepSS-DTIs’ (Bahi and Batouche, 2018) is based on deep semi-supervised learning, using large-scale chemical protein data to accurately predict potential new DTI. However, the input of these methods is the molecular fingerprint or molecular descriptor of the drug and the target, and the local connection between atoms and the local chemical structure of amino acids are not considered. The most advanced similarity-based methods use one-dimensional representations of drug and protein sequences, and use powerful advanced deep learning models to extract complex local chemical information and contextual relationships between local structures in the sequence (Abbasi et al., 2020; Chen et al., 2020; Huang et al., 2020; Karimi et al., 2019; Nguyen et al., 2020; Quan et al., 2019; Tsubaki et al., 2019). Finally, the feature information of the drug and the target is spliced and input into the neural network for prediction. Some of the methods listed above take the one-dimensional representation of the drug as a graph for feature representation. Other similar similarity-based methods (Lee et al., 2019; Torng and Altman, 2019; Zheng et al., 2020), although not completely using the one-dimensional sequence representation of the drug or target, are essentially the first feature extraction of the representation of the drug and target. After that, based on the assumption that compounds with similar structures should have similar properties, the extracted features of the drug and target are spliced and input into the neural network for DTI prediction. In general, the similarity-based method mainly intends to predict whether drug target pairs interact with each other through the characteristics of drug target pairs. The focus is on finding more advanced, better-performing and more reasonable feature extractors. The advantage of this type of method is that based on biological characteristics, the scalability is relatively strong, and in the case of strong generalization ability of the model, it can predict drug target interactions outside the dataset. However, this type of method also has some problems. It only considers the biological structure information of the drug target pair to predict their interaction, and the information considered is relatively single-sided. In addition, some models (Abbasi et al., 2020; Chen et al., 2020; Nguyen et al., 2020; Quan et al., 2019) show that the fixed protein length is set to about 1000, and the protein sequence length in the Drugbank database can be as long as 14 507. These

models cannot predict long-sequence drug target pairs outside the dataset (Fig. 1).

The network-based methods first construct a network containing drugs and targets and use graph-based technology to predict unknown edges based on known edges. The main idea is that drugs tend to bind to similar targets, and vice versa (Luo et al., 2017). An early method uses network-based inference (NBI) approach which only used drug–target bipartite network topology similarity to infer new targets for known drugs (Cheng et al., 2012). Ezzat et al. (2016) use the matrix factorization method of graph regularization to predict DTI with bipartite graphs as input. Recently, with more and more data sources related to drugs/targets, heterogeneous data sources can provide more information and different perspectives for predicting drug–target relevance. Studies have shown that the properties of drugs and targets can also be characterized by their various functional roles in biological systems (for example, protein–protein interactions and drug–disease associations) (Wan et al., 2019). Integrating various information from heterogeneous data sources can further improve the accuracy of DTI prediction (Luo et al., 2017). Liu et al. applied random walk with restart (RWR) to predict DTIs on the drug target heterogeneous network. Zeng et al. used a multimodal deep autoencoder to learn advanced features of drugs from heterogeneous networks to predict DTIs. In the past few years, the concepts, operations and models of graph neural networks have been constantly evolving and developing (Wu et al., 2020). And the graph convolutional neural network has shown strong performance in extracting embedded graphics (Schlichtkrull et al., 2017; Sun et al., 2020a,b; Zitnik et al., 2018), which greatly promotes the development of drug–target interaction prediction. Zhao et al. (2021) and Sun et al. (2020a,b) used a graph convolutional neural network to learn drugs and target heterogeneous network embedding. However, this type of method is mainly to characterize the nodes of the inherent heterogeneous network, and it is difficult to achieve target prediction for drugs outside the heterogeneous network. In addition, the feature representation of the node is mainly based on the topology information in the heterogeneous network, without in-depth consideration of the biological structure information of the drug and the target.

In this article, in order to solve the problems of similarity-based and network-based methods, we propose a new end-to-end method called ‘MultiDTI’ to predict new DTI from heterogeneous networks based on multi-modal representation. In the field of representation learning, the term ‘modality’ refers to a specific information encoding method or mechanism, and representation learning tasks involving multiple modalities will have multimodal characteristics (Guo et al., 2019). ‘MultiDTI’ uses our proposed joint representation framework based on heterogeneous networks to combine similarity-based methods with network-based methods. First, it takes one-dimensional sequence information of drugs and targets as input, and automatically learns sequence features with a long fixed input length through a low-complexity word-level deep convolutional neural network (Johnson and Zhang, 2017). Then, we propose a joint representation framework based on heterogeneous networks to project the representations of drugs, targets, side effects and diseases in a heterogeneous network into a common space. Finally, it predicts new DTI based on the distance between the drug and the target in the common space. Comparing ‘MultiDTI’ with similarity-based methods, it not only digs deeply into the structural information of drugs and targets, but also considers the interactions in the heterogeneous networks including drugs, targets, side effects and diseases. In addition, the model uses all the drug and target sequence information in the Drugbank database of version 5.1.7 to construct a vocabulary of sequence fragments, and sets the fixed sequence length of the model to the longest length of all sequences, so that the model can predict long-sequence drug–target pairs. Comparing ‘MultiDTI’ with network-based methods, it deeply mines the complex structural information of drugs and target sequences, and can predict potential targets for drugs outside of heterogeneous network datasets. The dataset of the model is based on a heterogeneous network, and the sequence information of all drugs and targets in the heterogeneous network is downloaded from the Drugbank database. The prediction performance of our end-to-end prediction model is significantly

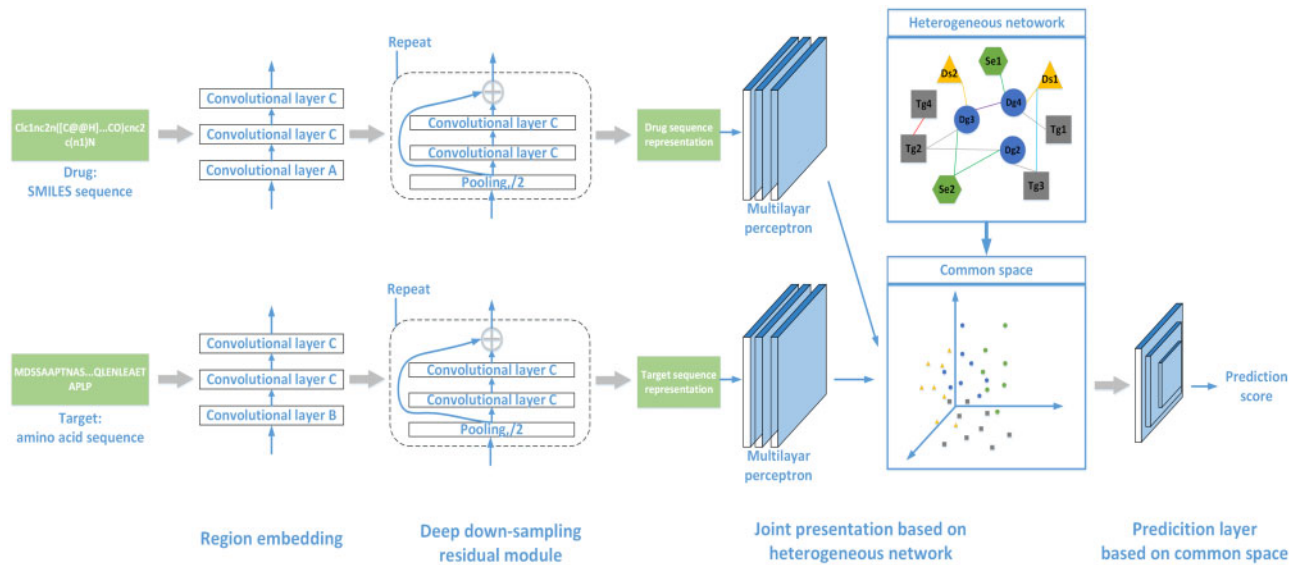


Fig. 1. The schematic diagram of ‘MultiDTI’. Our model takes the SMILE sequence of the drug, the amino acid sequence of the target, and the heterogeneous network they are in as input. First of all, SMILES sequence and amino acid sequence will enter the regional embedding module after n-gram embedding technology. In this module, we use a three-layer convolutional neural network to extract the regional embedding of each ‘word’ in the sequence. In the first convolutional layer, the embedding of each word contains not only its inherent information, but also the information of its neighbors on the left and right sides. After three layers of processing, the embedding of each word will contain several neighbor information nearby (see Section 2.3.1 for more details of the difference from previous work). The deep down-sampling residual module is composed of multiple down-sampling residual layers, each of which will reduce the internal representation in the length direction of the sequence by half until the internal representation in the length direction of the sequence becomes one. The sequence representation obtained in this way can contain long-distance correlation and rich global information (see Section 2.3.2 for more details on borrowing). In our joint representation learning framework, we take the correlation information in a heterogeneous network containing drugs, targets, side effects and diseases as constraints, and map the sequence representations of drugs and targets to the common space through a multilayer perceptron (see Section 2.4). In the final prediction layer based on the common space, we use the Euclidean distance to calculate the distance between the drug and the target, and map the distance between 0 and 1 through the neural network and the sigmoid activation function, and finally get the prediction score (see Section 2.5)

better than other models that use the same heterogeneous network dataset. In addition, some of the new DTI predicted by the ‘MultiDTI’ model have been verified by the latest version of the ChEMBL database, further demonstrating that our model has strong predictive capabilities. The above results show that ‘MultiDTI’ is a powerful and practical tool for predicting potential DTI. It can provide biologists with DTI candidates to reduce the workload of wet laboratory experiments and promote the development of new drug development and drug repositioning.

In short, the main contributions of this article are as follows:

- 1. We propose the first end-to-end deep learning model that combines two mainstream similarity-based methods and network-based methods to predict potential DTIs.
- 2. We proposed joint representation framework based on heterogeneous networks to make full use of the sequence structure information of the drug/target and the interaction or association information in the heterogeneous network containing drugs, targets, side effects and diseases.
- 3. The model supports the prediction of long-sequence DTI and can predict potential targets of drugs outside the dataset, which improves the applicability of the model.
- 4. Our model is superior to some state-of-the-art drug–target interaction prediction methods in evaluation indicators, and some of the predicted results are verified to be correct.

2 Materials and methods

2.1 Datasets

2.1.1 Heterogeneous network

We use a partial heterogeneous network dataset of a previous paper (Wan et al., 2019), which contains six related networks: drug–protein interaction network, drug–drug interaction network, protein–protein interaction network, drug–disease association network,

protein–disease association network and drug side-effect association network. Among them, the drug–target interaction network and the drug–drug interaction network are extracted from Drugbank Version 3.0 (Knox et al., 2011). The target–target interaction network is extracted from the HPRD database Release 9 (Keshava Prasad et al., 2009). The drug–disease association network and the target–disease association network are extracted from the Comparative Toxicogenomics Database (Davis et al., 2013). The drug–side effect association network is extracted from the SIDER database Version 2 (Kuhn et al., 2010). This heterogeneous network contains 708 drug nodes, 1512 target nodes, 5603 disease nodes and 4192 side effect nodes. It has a binary edge weight, that is, 1 represents a known interaction or association, and 0 represents an unknown or no interaction or association. More details can be found in Table 1.

2.1.2 Sequences of drug and target

The drugs in the heterogeneous network are represented by the SMILES sequence, and the targets in the heterogeneous network are represented by the original protein sequence. These sequences are mainly obtained from the Drugbank database through Drugbank ID and UniProt ID. The ones not found in the Drugbank database are downloaded from PubChem database and UniProt database.

Table 1. Nodes and edges in this heterogeneous network

Drugs	Targets	Diseases	Side effect
708	1512	5603	4192
Drug–Drug	Drug–target	Drug–disease	Drug–side effect
10 036	1923	199 214	80 164
	Target–target	Target–disease	
	7363	1 596 745	

2.2 Drug and target representation

We use n-gram embedding technology to define ‘words’ in protein sequences, that is, we divide these protein sequences into overlapping n-gram amino acids. Given a protein sequence $T = s_1, s_2, \dots, s_T$, where s_i is the i^{th} amino acid and T is the sequence length, the target can be expressed as:

$$T = (s_1, \dots, s_n), (s_2, \dots, s_{n+1}), \dots, (s_{T-n+1}, \dots, s_T) \quad (1)$$

where $(s_i, \dots, s_{i+n-1})$ represents the i^{th} word of the protein sequence. Because there are a total of 20 amino acids, the total number of words is 20^n . In order to control the vocabulary in an appropriate range and avoid low frequency words in the representation, we set n to 3.

In order to increase the applicability of the model, when we construct the protein vocabulary, we not only consider the protein sequences in the heterogeneous network dataset, but also consider all the protein sequences in the Drugbank database. Therefore, the maximum fixed length L_{max} of the protein sequence is considered to be 15 000, and shorter length proteins are padded with zeros. Finally, the embedded representation of the protein sequence is $T_e \in R^{L_{max} \times E_v}$ where E_v denotes the embedding vector of words in protein vocabulary.

The drug molecule is represented by the SMILES string containing a sequence of characters. Unlike protein sequence processing, $D = a_1, a_2, \dots, a_D$ are used to represent drugs where a_i denotes an atom or a structure indicator and D is the length of the drug sequence. According to the longest drug sequence length in the Drugbank database, we set the maximum fixed length l_{max} of the drug sequence to 1500, and shorter length drugs are padded with zeros. Finally, the embedded representation of the drug sequence is $D_e \in R^{l_{max} \times e_v}$ where e_v denotes the embedding vector of an atom or a structure indicator.

2.3 Deep convolutional neural network for long sequences

After obtaining the original embedding representation of the long sequence of the drug and the target, we use the deep pyramid CNN module (Johnson and Zhang, 2017) with lower computational complexity to effectively represent the long-range association in the long sequence and more global information.

2.3.1 Sequence region embedding

When we characterize a long sequence, we first need to characterize the sequence fragments. In short, let each word embedding not only contain its own information, but also the information of the neighbors on the left and right sides. We use a convolutional layer to learn this region embedding. This convolutional layer consists of multiple convolution kernels, each of which responds to a small region of each input. Given the original embedding representation of a sequence $S \in R^{l \times e_v}$, the calculation expressions of the n^{th} convolution kernel and the i^{th} region are as follows:

$$S'_{i,n} = W_n \odot S_{i,1} + b_n \quad (2)$$

where $S_{i,1} \in R^{3 \times e_v}$ denotes the i^{th} region representation in the 1th channel of the original embedding representation S . Here, we fix the number of convolution kernels to N , and the size of the kernel in the sequence direction (the size of the small area covered) is fixed to 3. So, the i^{th} region here is a region with a range of 3 composed of the i^{th} word with left and right neighbors. $W_n \in R^{3 \times e_v}$ and $b_n \in R^1$ respectively represent the weight and bias of the n^{th} convolution kernel. Then after the convolutional layer processing with n convolution kernels, the representation of the sequence becomes $S' \in R^{(l-2) \times n}$.

Next, we use a two-layer convolutional layer with the same number of input and output channels to allow the region embedding to interact with its neighbors. Unlike the original deep pyramid CNN module, we do not use the shortcut connection module here, and pay more attention to the embedding of the region. Given

embedding representation of a sequence with a zero added to the beginning and the end $S' \in R^{l \times n}$, the calculation expressions of the m^{th} convolution kernel and the i^{th} region are as follows:

$$S''_{i,m} = \sum_{i=1}^n (W_m \odot ReLU(S'_{i,n}) + b_m) \quad (3)$$

where $S'_{i,n} \in R^{3 \times 1}$ denotes the i^{th} region representation in the n^{th} channel of S' . $W_m \in R^{3 \times 1}$ and $b_n \in R^1$ respectively represent the weight and bias of the m^{th} convolution kernel. $ReLU(\cdot)$ denotes a non-linear activation function and pre-activation is used here. After two identical non-linear convolutional layers are processed, the sequence is expressed as $S'' \in R^{(l-2) \times n}$.

2.3.2 Deep down-sampling residual module

In this step, we use down-sampling with the number of feature maps fixed and shortcut connections with pre-activation to give long-range associations to long sequences, thereby obtaining more global information. The deep down-sampling residual module is composed of multiple down-sampling residual layers, and each down-sampling residual layer reduces the internal representation of the sequence in the length direction by half. After multiple down-sampling residual layer processing, the internal representation of the sequence in the length direction will become 1. Given the sequence representation $S^{(i)}$ obtained by the previous layer, the result of down-sampling residual layer processing is expressed as follows:

$$S^{(i+1)} = \text{down}(S^{(i)}) + f(\text{down}(S^{(i)})) \quad (4)$$

where $S^{(0)} = S''$. $\text{down}(\cdot)$ denotes max-pooling function with size three and stride two. That is, it generates a new embedding representation of the sequence by embedding the maximum value in three consecutive regions with a stride of two in the length direction of the sequence. $f(\cdot)$ denotes the two-layer convolutional layer corresponding to formula 3.

The method of first non-linear activation and then linear weighting in the function $f(\cdot)$ makes the result of formula 4 including addition also the result of linear weighting. Intuitively, this ‘linear’ simplifies the training of the deep network. In addition, when the number of feature maps is fixed, each time sequence embedding is processed by a down-sampling residual layer, the calculation time is halved as the sequence embedding size is halved. The total calculation time of the deep down-sampling residual module is limited to a constant range—the time taken is twice the calculation of $S^{(0)}$, which makes it very suitable for long sequence characterization. Finally, we complete the feature extraction of the sequence, which is expressed as $S^{fin} \in R^n$.

2.4 Joint presentation based on heterogeneous network

After obtaining the final representation of the drug and the target, most of the similarity-based methods mentioned before are to concatenate the representation of the drug and the target and input it into the neural network for DTI prediction. However, this simple combination strategy ignores the key correlation between the drug and the target. Inspired by multi-modal representation learning (Cao et al., 2020; Guo et al., 2019; Tadas et al., 2019), we propose a joint representation framework based on heterogeneous networks to project the representations of drugs, targets, side effects and diseases in a heterogeneous network into a common space. Two nodes connected by an edge in a heterogeneous network will be closer in the common space. This method considers the correlation between multiple modalities to obtain a better embedding representation, which is particularly important for the subsequent interactive learning process.

In the common space, the vector representations of drugs, targets, side effects and diseases are forced to be linked under the constraints of modal correlation. Side effects and diseases have no biological characteristics, so we first assume that the representation of side effects and diseases in common space obeys Gaussian distribution:

$$\begin{cases} E_i^{se} \sim N(0, \sigma^2) \\ E_j^{ds} \sim N(0, \sigma^2) \end{cases} \quad (5)$$

where E_i^{se} represents the learnable embedding of side effect i in common space. E_j^{ds} represents the learnable embedding of disease j in common space. In our implementation, σ^2 is set to 0.01.

In order to project the representation of the drug and target sequence into the common space, we used the following two parallel mapping functions:

$$\begin{cases} E_i^{dg} = f_{dg}(S_i^{dg}) \\ E_j^{tg} = f_{tg}(S_j^{tg}) \end{cases} \quad (6)$$

where S_i^{dg} and S_j^{tg} respectively represent the sequence representation of the drug i and the sequence representation of the target j . E_i^{dg} and E_j^{tg} respectively represent the vector representation of the drug and the target in the common space. $f_{dg}(\cdot)$ and $f_{tg}(\cdot)$ are independent mapping functions approximated by a neural network with a multi-layer perceptron structure. Here, taking drugs as an example, the more similar the sequence structure is, the closer they will be in the public space.

In order to regularize the learnable embedding of side effects and diseases in public spaces and regularize learnable mapping functions, we propose a joint representation framework based on heterogeneous networks. The objective function of the joint representation based on heterogeneous networks is as follows:

$$L_{com} = \sum_{r \in R} \sum_{(M_i, N_j, r) \in E} \|E_i^M - E_j^N\|^2 \quad (7)$$

where the node type set $V = \{\text{drug, target, disease, side effect}\}$. The relation type set $R = \{\text{drug-drug-interaction, drug-protein-interaction, drug-disease-association, drug-side effect-association, protein-protein-interaction, protein-disease-association}\}$. E represents the set of edges in a heterogeneous network. M_i and N_j respectively represents node i of type M and node j of type N . E_i^M and E_j^N represent the embedding of two nodes M_i and N_j with relation type r in the common space.

As such, drug embedding, target embedding, side effect embedding and disease embedding in the common space can be mutually enhanced through the correlation in the heterogeneous network, so as to obtain better characterization. It is worth mentioning that their embedding in the common space is obtained through the representation of biological sequences and interaction in heterogeneous networks, which is suitable for cold start scenarios.

2.5 Prediction layer based on common space

As mentioned earlier, the distance between two nodes connected by edges in a heterogeneous network will be closer in the common space. Therefore, we designed a prediction layer to convert the distance between the drug and the target in the common space into the prediction score of DTI. Given a drug and a target, their embeddings in the common space are expressed as E_i^{dg} and E_j^{tg} respectively, then the prediction score between them is expressed as follows:

$$x_{ij} = \text{Sigmoid}\left(BN\left(W E_i^{dg} - E_j^{tg} 2 + b\right)\right) \quad (8)$$

where W and b respectively represent the weight and bias of prediction layer. $\text{Sigmoid}(\cdot)$ represents S-type saturated non-linear activation function. $BN(\cdot)$ represents batch normalization operation. It is placed in front of the sigmoid function to prevent the data distribution from shifting and entering the saturated area, thereby avoiding the disappearance of the gradient. The prediction score is regularized by sigmoid function to a range of 0 to 1.

We treat the task of predicting DTI as a binary classification problem, that is, the interaction between the drug and the target is assigned a value of 0 or 1. We use the binary cross-entropy function to fit the prediction score and the label value. The objective function is as follows:

$$\begin{aligned} L_{pred} &= - \sum_{(i,j) \in Z^+} \log_e(x_{ij}) - \sum_{(i,j) \in Z^-} \log_e(1 - x_{ij}) \\ &= - \sum_{(i,j) \in Z^+ \cup Z^-} y_{ij} \log_e(x_{ij}) + (1 - y_{ij}) \log_e(1 - x_{ij}) \end{aligned} \quad (9)$$

where Z^+ represents the set of observed interactions of drugs and targets. Z^- represents the set of negative interactions of drugs and targets. x_{ij} and y_{ij} respectively represent the prediction score and label value between drug i and target j .

In order to regularize the parameters in the 'MultiDTI' we proposed, we finally define our objective function by jointly regularizing joint representation based on heterogeneous network and the prediction layer based on the common space:

$$L_{final} = L_{com} + L_{pred} \quad (10)$$

2.6 Training and testing

In the heterogeneous network dataset, there are fewer positive samples of drug-target pairs than negative samples. In order to increase the generalization ability of the model, we oversample the positive samples by 10 times and under-sample the negative samples to make the size the same as the number of positive samples after 10 times of oversampling. After that, we perform 10-fold cross-validation on the sampled data, that is, for each fold, a randomly selected 90% positive and negative sample pairs are used as training data, and the remaining 10% is used to test the prediction performance of the model. The final performance of the model takes the average of the 10-fold results. Here, we use mini-batch gradient descent to train our model, and use the backpropagation strategy to update the corresponding parameters of the model. Specifically, in each epoch, each batch of training data takes a gradient step to optimize the objective function. In our implementation, we first optimize L_{com} with several batches, and then optimize L_{pred} with several batches.

To prevent the neural network in our model from overfitting the training data, we use a method called Dropout. This method stops some neurons from working during the training process and only updates the parameters of those neurons that have not stopped working. In our implementation, dropout is used in the neural network where the mapping function in Equation 6 is located. In this way, our model will have better generalization ability and can be extended to test scenarios. In addition, dropout is only used in the training phase and needs to be disabled during the testing phase.

2.7 Performance test metrics

In this work, we used six performance indicators: accuracy, precision, recall, F1 score, area under the receiver operating characteristic curve (AUC), and area under the precision recall curve (AUPRC) to evaluate the model. AUC and AUPRC are obtained based on the area under the curve, and the other four indicators are defined as follows:

$$\text{Accuracy} = \frac{TP + FN}{TP + FP + TN + FN} \quad (11)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

$$\text{F1score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (14)$$

where TP (True Positive) and TN (True Negative) respectively represent the numbers of correctly predicted positive and negative samples. FP (False Positive) and FN (False Negative) respectively represent the numbers of wrong predicted positive and negative samples. Specifically, if the predicted score of a certain drug-target pair is greater than 0.5, the model determines that the drug and the target interact, which is a predicted positive sample, and vice versa.

3 Results

3.1 MultiDTI has superior predictive ability

We compare our proposed model ‘MultiDTI’ with three other models using similar datasets: ‘HNM’ (Wenhui *et al.*, 2014), ‘DTINet’ (Luo *et al.*, 2017) and ‘NeoDTI’ (Wan *et al.*, 2019). They use different methods to learn the feature representations of nodes in heterogeneous networks, and then make DTI predictions based on the feature representations. On the dataset, we use the same heterogeneous network dataset. The difference is that we delete the drug structure similarity network and the protein sequence similarity network in the original heterogeneous network (Wan *et al.*, 2019), and use the sequence feature extractor to deeply extract the sequence characteristics of the drug and the target in the heterogeneous network. It is worth noting that we use the remaining 10% of a partial heterogeneous network dataset (Section 2.1.1) divided by the 10-fold cross-validation method (Section 2.6) to test the performance of the model. From Table 2 we can see that our proposed model ‘MultiDTI’ is the highest among indicators such as Recal, F1 score, AUC and AUPRC. Specifically, the other four indicators of the ‘HNM’ model are not available, and we use ‘NA’ to represent them in the table. Accuracy is the most common evaluation indicator, which is the number of samples that are predicted correctly divided by the number of all samples. In the case of unbalanced positive and negative samples, the evaluation index of accuracy has a big flaw. ‘NeoDTI’ characterizes heterogeneous networks and trains the model by reconstructing the network. The number of negative samples used for training and testing is ten times the number of positive samples. Because there are too many negative samples during the test, the ‘NeoDTI’ model tends to predict the samples as negative samples, and the accuracy value will be higher. Precision represents the proportion of samples classified as positive examples that are actually positive examples. Recall represents the proportion of all samples that are actually positive that are predicted to be positive. If the recall is relatively low, it means that more positive samples are predicted by the model as negative samples. The F1 score comprehensively measures the accuracy and recall of the model. It has a good reputation and objectivity. From this indicator, our model is better than other models, and it is 0.06 higher than the second highest model. At the same time, AUC and AUPRC are commonly used evaluation indicators to test the capabilities of binary classifiers. Our model is also the best in these two indicators. In general, ‘MultiDTI’ deeply mines the sequence information of drugs and targets, combines multiple correlation information in heterogeneous networks, and has a powerful ability to predict new DTI.

3.2 More correlation information and over-sampling can improve prediction performance

Here, we first evaluate the impact of heterogeneous network datasets on the performance of the model. While keeping other parameters unchanged, we only use the drug–target interaction network to retrain the model and obtain performance indicators. It can be seen from Figure 2 that only using drug–target interaction network, the model’s performance indicators AUC and AUPRC have decreased to varying degrees, and the response on the AUPRC indicator is more obvious. These results indicate that the abundant relevant information in heterogeneous networks enables drugs and targets to be better expressed in public spaces, thereby improving the performance

Table 2. Comparison with other models on heterogeneous network dataset

Models	Acc	Pre	Rec	F1	AUC	AUPRC
Our model	0.873	0.876	0.882	0.868	0.961	0.947
NeoDTI	0.969	0.928	0.716	0.808	0.957	0.874
DTINet	0.524	1.0	0.048	0.090	0.914	0.932
HNM	NA	NA	NA	NA	0.890	0.572

Note: Bold entries indicate that the value is the largest in the column to which it belongs.

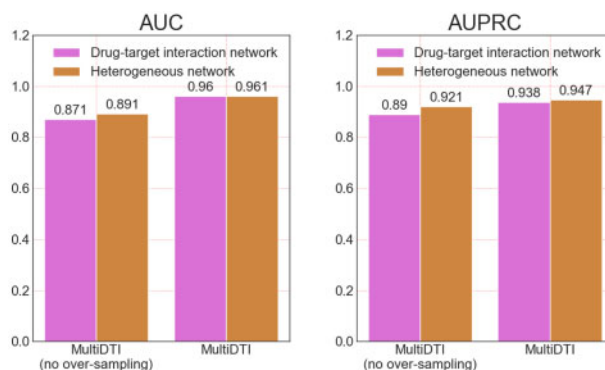


Fig. 2. Whether the model is over-sampling, whether the model uses heterogeneous network for performance comparison. The evaluation indicators used are AUC and AUPRC

of predicting DTI. On the other hand, this also proves the superiority of our proposed joint representation learning framework that combines the correlation information of heterogeneous networks and sequence biological structure information.

In the heterogeneous network, there is an imbalance between the positive and negative samples of the drug–target pair, and the positive samples are much less than the negative samples. Based on this fact, when we construct the dataset, we oversample the positive samples by 10 times and under-sample the negative samples. The number of negative samples after under-sampling is the same as the positive samples after positive sampling. Under the condition of ensuring that other parameters are the same, we compare the model with positive sample oversampling and without positive sample oversampling, and the result is shown in Figure 2. We found that the model with 10 times oversampling of positive samples has significant improvement in AUC and AUPRC indicators. This shows that oversampling of positive samples (full use of known drug–target pairs) can enhance the generalization ability of the model and reduce the negative impact of data imbalance in heterogeneous networks.

3.3 Some of the predicted new DTI was confirmed by the ChEMBL database

ChEMBL is a biologically active drug-like small molecule database, which contains two-dimensional chemical structures, computational properties and biological activities. The data is extracted and collated from the main scientific literature and covers a large part of SAR and modern drug discovery (Anna *et al.*, 2017). We use this database to verify the new drug target pairs predicted by our model. Here, we use a complete heterogeneous network containing drugs, targets, side effects and diseases as a dataset, and use positive samples with 10 times oversampling and the same number of under-sampled negative samples to train ‘MultiDTI’ to predict new DTIs. Through the 10-fold cross-validation method, we have obtained ten trained models, and we will use the first trained model for the prediction of the new DTI. We pair all drugs and all targets in the dataset (excluding drug–target pairs that have interactions in the dataset), input their sequences into the model and get the corresponding prediction scores. We sort the predicted scores from largest to smallest, and list the top 20 drug targets in the predicted scores in Table 3 (for the top 1000 predicted results, please see Supplementary Data). As can be seen from Table 3, the ChEMBL database has verified that 3 predicted drug–targets pairs. Those drug target pairs that have not been verified are good candidates for wet experiment exploration. Overall, these new DTIs predicted by ‘MultiDTI’ and confirmed by the ChEMBL database further prove the powerful predictive ability of our model.

3.4 Target prediction for new chemical entities

Here we select some new drugs to test the effectiveness of our model connecting new chemical entities with known DTI heterogeneous

Table 3. Top 20 prediction scores of the 'MultiDTI' model

Drug name	Protein name	Prediction score	ChEMBL verification
Olopatadine	Glucocorticoid receptor	0.8216	Unproved
Halothane	Glucocorticoid receptor	0.8211	Unproved
Zonisamide	Glucocorticoid receptor	0.8156	Unproved
Sunitinib	Glucocorticoid receptor	0.8080	Unproved
Olopatadine	Mu-type opioid receptor	0.8062	Unproved
Halothane	Mu-type opioid receptor	0.8048	Unproved
Amiloride	Glucocorticoid receptor	0.8047	Proved
Olopatadine	Prostaglandin G/H synthase 1	0.8043	Unproved
Halothane	Prostaglandin G/H synthase 1	0.8038	Unproved
Arsenic trioxide	Glucocorticoid receptor	0.8022	Unproved
Olopatadine	Prostaglandin G/H synthase 2	0.8011	Unproved
Halothane	Prostaglandin G/H synthase 2	0.8008	Unproved
Zonisamide	Mu-type opioid receptor	0.8003	Unproved
Vitamin A	Glucocorticoid receptor	0.8001	Proved
Olopatadine	Beta-2 adrenergic receptor	0.8001	Unproved
Halothane	Beta-2 adrenergic receptor	0.7997	Unproved
Dasatinib	Glucocorticoid receptor	0.7992	Unproved
Zonisamide	Prostaglandin G/H synthase 1	0.7984	Unproved
Hydroxocobalamin	Glucocorticoid receptor	0.7978	Unproved
Cladribine	Glucocorticoid receptor	0.7959	Proved

Note: This table lists the top 20 drug target pairs with the highest predicted scores and their verification in the ChEMBL database.

networks. First, we input a medicine outside the heterogeneous network into the trained model. Second, the model uses a sequence feature extractor to characterize the drug according to its sequence. Third, the model uses a multilayer perceptron to map the sequence representation of the drug to the trained public space, and finally the model uses a prediction layer based on the public space to predict the target of the drug. After obtaining the predicted targets of these drugs, we still use the ChEMBL database to verify the predicted results. We input Cyclosporine, a new drug entity with a longer sequence length, into the model, and the results verified by the ChEMBL database are among the top 10 targets with predicted scores, and 9 targets are verified to interact with Cyclosporine. In addition, we input Glyburide, a new drug entity with a shorter sequence length, into the model, and the verification result obtained is that among the top 10 targets with predicted scores, 9 targets have been verified to interact with Glyburide. These facts indicate that our model has strong performance in target prediction of new drug entities.

4 Conclusion

In this work, we proposed a multi-modal DTI prediction model named 'MultiDTI' based on heterogeneous networks and sequences. Our model uses joint learning in multimodal representation learning to combine similarity-based methods with network-based methods. It not only deeply mines the structural information of drugs and targets, but also considers the interaction or association information in heterogeneous networks. Our model solves the problem that most network-based methods can only predict drug targets in the network dataset, and our model can support the prediction of drug-target pairs outside the dataset. In addition, our model constructs a vocabulary of sequence fragments from the sequences of all drugs and targets in the DrugBank database, and sets the fixed sequence length of the model to the longest length of all sequences which can support the prediction of most long-sequence drug-target pairs. Compared with other models, our model has a significant improvement in prediction performance in experiments, and some predicted new DTIs have been confirmed. Therefore, we believe that 'MultiDTI' is a powerful and practical tool for predicting new DTI, which can promote the development of drug discovery or drug repositioning.

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